Series : A3BAB/1

SET-2

प्रश्न-पत्र कोड
Q.P. Code **55/1/2**

रोल नं.
Roll No.

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें ।
Candidates must write the Q.P. Code on the title page of the answer-book.

नोट

(I) कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ **12** हैं ।

(II) प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को छात्र उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें ।

(III) कृपया जाँच कर लें कि इस प्रश्न-पत्र में **12** प्रश्न हैं ।

(IV) **कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, प्रश्न का क्रमांक अवश्य लिखें ।**

(V) इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है । प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा । 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे ।

NOTE

(I) Please check that this question paper contains **12** printed pages.

(II) Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.

(III) Please check that this question paper contains **12** questions.

(IV) **Please write down the Serial Number of the question in the answer-book before attempting it.**

(V) 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the candidates will read the question paper only and will not write any answer on the answer-book during this period. *

भौतिक विज्ञान (सैद्धान्तिक)

PHYSICS (Theory)

निर्धारित समय : 2 घण्टे | *अधिकतम अंक : 35*

Time allowed : 2 hours | *Maximum Marks : 35*

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) *This question paper contains 12 questions.* ***All*** *questions are compulsory.*

(ii) *This question paper is divided into three sections – Section A, B and C.*

(iii) ***Section A :*** *Q. Nos.* ***1*** *to* ***3*** *are of* ***2*** *marks each.*

(iv) ***Section B :*** *Q. Nos.* ***4*** *to* ***11*** *are of* ***3*** *marks each.*

(v) ***Section C :*** *Q. Nos.* ***12*** *is a case study based question of* ***5*** *marks.*

(vi) *There is no overall choice in the question paper. However, internal choice has been provided in some of the questions. Attempt any one of the alternatives in such questions.*

(vii) *Use of log tables is permitted, if necessary, but use of calculator is not permitted.*

$c = 3 \times 10^{8}$ m/s

$h = 6.63 \times 10^{-34}$ Js

$e = 1.6 \times 10^{-19}$ C

$\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1}

$\varepsilon_0 = 8.854 \times 10^{-12}$ C^2 N^{-1} m^{-2}

$\frac{1}{4\pi\varepsilon_0} = 9 \times 10^{9}$ N m^2 C^{-2}

Mass of electron (m_e) = 9.1×10^{-31} kg

Mass of neutron = 1.675×10^{-27} kg

Mass of proton = 1.673×10^{-27} kg

Avogadro's number = 6.023×10^{23} per gram mole

Boltzmann constant = 1.38×10^{-23} JK^{-1}

SECTION – A

1. Explain the formation of depletion layer and barrier potential in a p-n junction diode. **2**

2. Draw energy band diagrams of n-type and p-type semiconductors at temperature T > 0K, depicting the donor and acceptor energy levels. Mention the significance of these levels. **2**

3. (a) Draw the graph showing the variation of the number (N) of scattered alpha particles with scattering angle (θ) in Geiger – Marsden experiment. Infer two conclusions from the graph. **2**

OR

(b) Plot suitable graphs to show the variation of photoelectric current with the collector plate potential for the incident radiation of

(i) the same intensity but different frequencies ν_1, ν_2 and ν_3 ($\nu_1 < \nu_2 < \nu_3$)

(ii) the same frequency but different intensities I_1, I_2 and I_3 ($I_1 < I_2 < I_3$)

SECTION – B

4. Explain with the help of a suitable diagram, the phenomenon on which an optical fibre works. Mention any two uses of optical fibres. **3**

5. Ultra-violet light of wavelength 200 nm from a source is incident on a metal surface. If the stopping potential is –2.5 V, (a) calculate the work function of the metal, and (b) How would the surface respond to a high intensity red light of wavelength 6328 Å produced by a laser ? **3**

6. (a) A parallel beam of light of wavelength 600 nm is incident normally on a slit of width 0.2 mm. If the resulting diffraction pattern is observed on a screen 1 m away, find the distance of **3**

(i) first minimum, and

(ii) second maximum, from the central maximum.

OR

(b) A thin equiconvex lens of radius of curvature R made of material of refractive index μ_1 is kept coaxially, in contact with an equiconcave lens of the same radius of curvature and refractive index μ_2 $(>\mu_1)$. Find :

(i) the ratio of their powers, and

(ii) the power of the combination and its nature.

7. (a) (i) Monochromatic light is incident on a surface separating two media. The frequency of the light after refraction remains unaffected but its wavelength changes. Why ? **3**

(ii) The frequency of an electromagnetic radiation is 1.0×10^{11} Hz. Identify the radiation and mention its two uses.

OR

(b) (i) Trace the path of a ray of light PQ which is incident at an angle i on one face of a glass prism of angle A. It then emerges out from the other face at an angle e. Use the ray diagram to prove that the angle through which the ray is deviated is given by $\angle\delta = \angle i + \angle e - \angle A$.

(ii) What will be the minimum value of δ if the ray passes symmetrically through the prism ?

8. State the working principle of an LED. Write any two important advantages and two disadvantages of LED. **3**

9. (a) Sketch the energy level diagram for hydrogen atom. **3**

(b) Find the ratio of the longest and the shortest wavelength in Lyman series in hydrogen atom.

10. Find the two possible positions of an object kept in front of a lens of + 5.0 D, so that the image formed in both cases is four times magnified. **3**

11. Calculate the energy released in MeV in the following reaction : **3**

$${}_{1}^{2}H + {}_{1}^{3}H \longrightarrow {}_{2}^{4}He + n$$

Given : $m\left({}_{1}^{2}H\right) = 2.014102$ u

$m\left({}_{1}^{3}H\right) = 3.016049$ u

$m\left({}_{2}^{4}He\right) = 4.002603$ u

$m_n = 1.008665$ u

SECTION – C
CASE STUDY

12. The principle of superposition is used to understand the phenomenon of interference of light waves. The principle states that at a particular point, the resultant displacement produced by a number of waves is the vector sum of the displacements produced by each wave. Light waves from two coherent sources produce interference pattern. Thomas Young devised a way to obtain two coherent sources using two identical pinholes (S_1 and S_2) illuminated by a single monochromatic pinhole source S. Using these sources in his experiment known as Young's double slit experiment, Young studied the interference pattern. The pattern consists of alternate bright and dark fringes. The distance between two successive bright or dark finges depends on the distance between S_1 and S_2, the distance of the screen from the plane of S_1S_2 and the wavelength of light used. **5**

I. Consider the following waves :

(i) $y_1 = a \sin \omega t$

(ii) $y_2 = a \sin 2\omega t$

(iii) $y_3 = a \sin (2\omega t + \phi)$

(iv) $y_4 = a \sin \left(4\omega t + \frac{\pi}{2}\right)$

Which pair of the waves coming from two sources S_1 and S_2 will produce interference ?

(A) (i) and (ii) (B) (ii) and (iii)

(C) (iii) and (iv) (D) (iv) and (i)

II. Two light waves of the same intensity I_0 each, having a path difference of $\lambda/4$, emanating from two coherent sources, meet at a point. The resultant intensity at the point will be

(A) Zero (B) I_0

(C) $2\,I_0$ (D) $4\,I_0$

III. Vandana performs Young's double slit experiment by using orange, green and red lights successively. If the fringe widths measured in the three cases are ω_1, ω_2 and ω_3 respectively, then which of the following is correct ?

(A) $\omega_2 > \omega_1 > \omega_3$ (B) $\omega_1 > \omega_2 > \omega_3$

(C) $\omega_2 > \omega_3 > \omega_1$ (D) $\omega_3 > \omega_1 > \omega_2$

IV. In a Young's double slit experiment, the slit separation is 0.8 mm and the interference pattern is obtained on a screen kept 50 cm from the plane of the slits S_1 and S_2. If the first bright fringe is formed 0.4 mm from the central maximum, the wavelength of light used is

(A) 480 nm (B) 560 nm

(C) 640 nm (D) 680 nm

V. Consider the effect on the angular separation of the fringes in a Young's double slit experiment due to the following operations :

(i) the screen is moved away from the plane of the slits,

(ii) the separation between the two slits is increased till fringes are observed.

Which of the following options is correct ?

(A) It remains constant in both cases.

(B) It decreases in both cases.

(C) It remains constant in (i) but decreases in (ii).

(D) It decreases in (i) but remains constant in (ii).